Technical Manual

Printing & Food System

Version 1.0 · October 2026

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Printing & Food System (*Sistema de Impresiones y Comedor*) · Version 1.0

For technical staff who install, maintain or modify the system.

> Source code identifiers (files, tables, fields, constants) are in Spanish. English meanings are given where useful.

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1. Overview

Web application for **order and sales management** in a business with two areas: **Impresiones** (Printing: prints, copies and services) and **Comida** (Food: lunches, breakfasts, snacks and drinks). One order can mix items from both areas.

Modules:

Module	Files
Authentication	`login.php`, `logout.php`, `clave.php` (change password)
Dashboard (today's summary)	`index.php`
Orders	`pedido_nuevo.php` (new), `pedidos.php` (list), `pedido_ver.php` (view), `pedido_estado.php` (status)
Catalog	`catalogo.php`, `catalogo_form.php`, `catalogo_eliminar.php`
Reports	`reportes.php`

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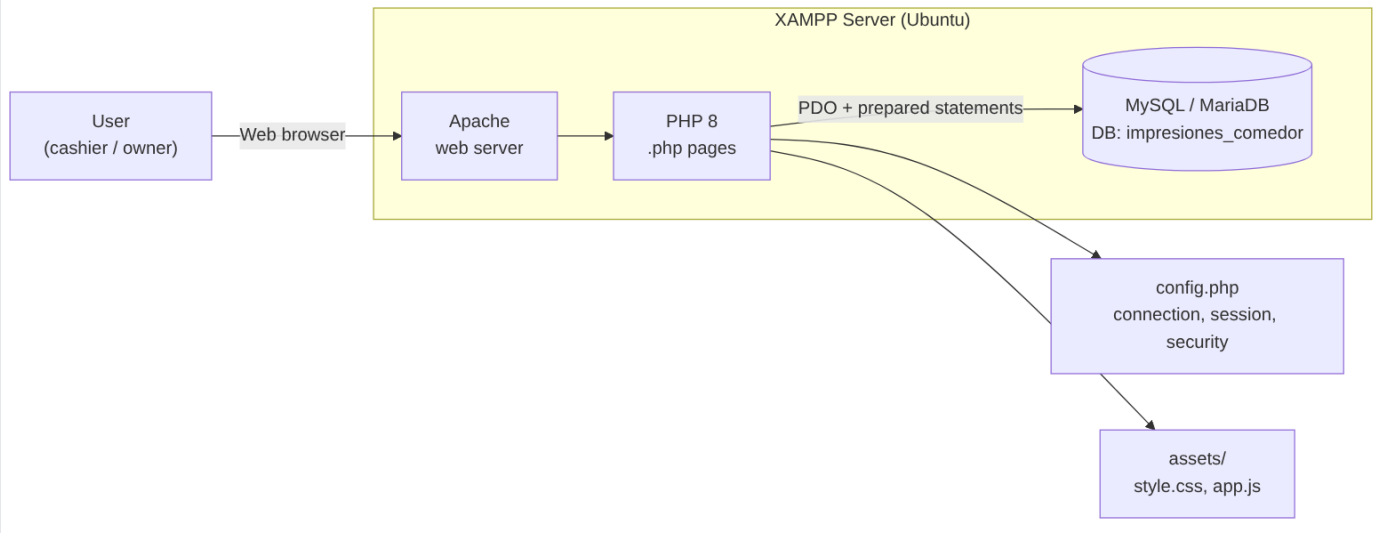
2. Technology stack

Layer	Technology
Web server	Apache 2.4 (bundled with XAMPP)
Server language	PHP 8.x, no framework
Database	MySQL 8 / MariaDB 10 (bundled with XAMPP)
Data access	PDO with prepared statements
Front end	HTML5, responsive CSS3 and plain JavaScript
Version control	Git / GitHub

No Composer, npm or external libraries are required.

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3. Architecture



It is a **3-tier client-server** architecture:

1. **Presentation:** the browser renders the HTML generated by PHP, styled by `assets/style.css`, with `assets/app.js` for interaction.
2. **Logic:** each `.php` file handles the request, validates input and applies the business rules.
3. **Data:** MySQL stores the information. PHP connects through PDO, configured in `config.php`.

Every page follows the same pattern:

```
require 'config.php';   → connection, session and helper functions
requiere_login();       → redirect to login.php if there is no session
[handle POST]           → check CSRF, validate, save, redirect
[query data]            → SELECT with prepared statements
require 'header.php';   → menu
[page HTML]
require 'footer.php';   → footer and JavaScript
```

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4. File structure

```
impresiones/
├── config.php              Settings, PDO connection, session, CSRF and helpers
├── header.php / footer.php Shared layout (menu and footer)
├── login.php / logout.php  Log in / log out
├── index.php               Dashboard: today's sales per area and active orders
├── pedido_nuevo.php        New order form (uses a transaction)
├── pedidos.php             Order list with filters (date, status, text)
├── pedido_ver.php          Order details, printable ticket, status buttons
├── pedido_estado.php       Changes status (POST only)
├── catalogo.php            Catalog list with filters
├── catalogo_form.php       Create / edit an item
├── catalogo_eliminar.php   Deletes an item, or deactivates it if it has sales
├── reportes.php            Reports for a date range
├── clave.php               Change password
├── assets/
│   ├── style.css           Styles (including @media print for the ticket)
│   └── app.js              Live order total
├── database/
│   ├── database.sql        Database creation script with seed data
│   └── .htaccess           Blocks web access to this folder
├── docs/                   Manuals and diagrams (ES / EN)
├── .htaccess               Apache hardening (no directory listing; blocks .sql and .md)
├── .gitignore
├── LICENSE                 MIT license
└── README.md
```

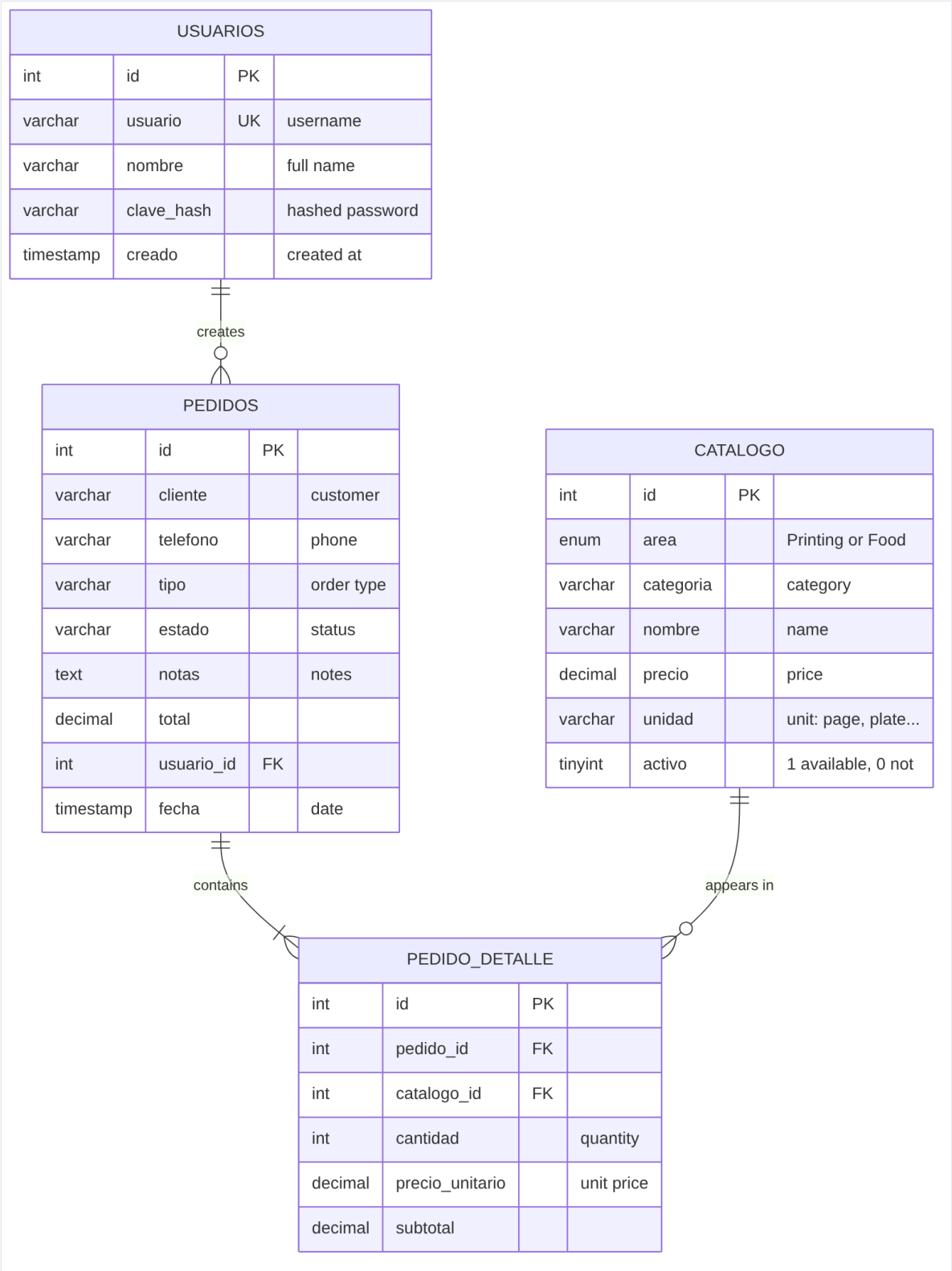
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5. Database

- **Name:** `impresiones_comedor`
- **Engine:** InnoDB · **Charset:** `utf8mb4_unicode_ci`

- **Script:** `database/database.sql`

5.1 Entity-Relationship Diagram



5.2 Relationships

Relationship	Type	On delete
`usuarios` → `pedidos`	One to many	`ON DELETE SET NULL` (order is kept)
`pedidos` → `pedido_detalle`	One to many	`ON DELETE CASCADE`
`catalogo` → `pedido_detalle`	One to many	`ON DELETE RESTRICT` (items already sold cannot be deleted)

5.3 Data dictionary

Table `usuarios` (users): people who can log in.

Field	Type	Null	Description
id	INT AUTO_INCREMENT	No	Primary key
usuario	VARCHAR(50) UNIQUE	No	Login username
nombre	VARCHAR(100)	No	Full name (printed on the ticket as "Atendió", served by)
clave_hash	VARCHAR(255)	No	Password hashed with `password_hash()` (bcrypt)
creado	TIMESTAMP	No	Creation date

Table `catalogo` (catalog): printing services and food items.

Field	Type	Null	Description
id	INT AUTO_INCREMENT	No	Primary key
area	ENUM('Impresiones','Comida')	No	Business area
categoria	VARCHAR(50)	No	Category (Copies, Lunches, Drinks…)
nombre	VARCHAR(100)	No	Service or dish name
precio	DECIMAL(10,2)	No	Current unit price
unidad	VARCHAR(30)	No	Unit charged (page, plate…)
activo	TINYINT(1)	No	1 = available, 0 = unavailable

Table `pedidos` (orders): order header.

Field	Type	Null	Description
id	INT AUTO_INCREMENT	No	Primary key (order number)
cliente	VARCHAR(100)	No	Customer name
telefono	VARCHAR(20)	Yes	Customer phone
tipo	VARCHAR(30)	No	To go / Eat here / Pick up later
estado	VARCHAR(20)	No	Pendiente / En proceso / Listo / Entregado / Cancelado
notas	TEXT	Yes	Special instructions
total	DECIMAL(10,2)	No	Sum of subtotals
usuario_id	INT (FK)	Yes	User who registered the order
fecha	TIMESTAMP	No	Order date and time

Table `pedido_detalle` (order details): order lines.

Field	Type	Null	Description
id	INT AUTO_INCREMENT	No	Primary key
pedido_id	INT (FK)	No	Parent order
catalogo_id	INT (FK)	No	Catalog item sold
cantidad	INT	No	Quantity
precio_unitario	DECIMAL(10,2)	No	**Price at the time of sale** (historical)
subtotal	DECIMAL(10,2)	No	quantity × unit price

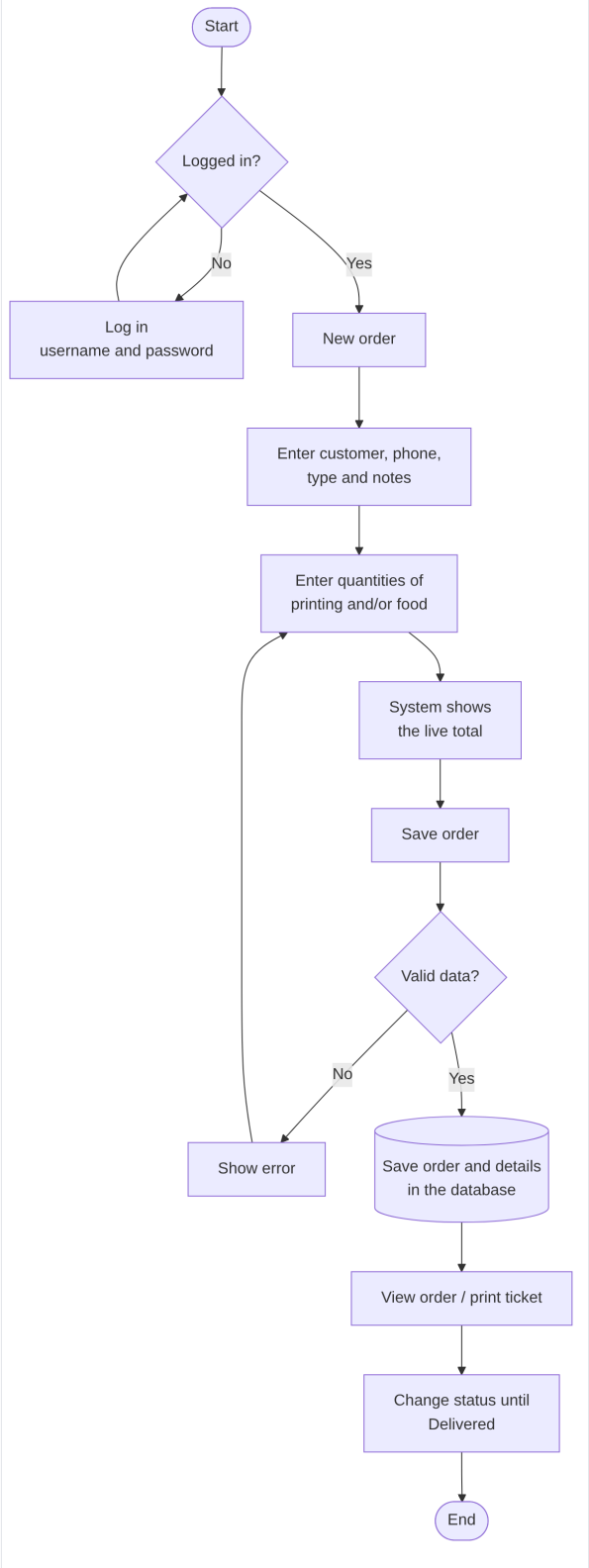
> The price is copied into `pedido_detalle.precio_unitario` when the sale is made. If the catalog price changes later, past orders and reports stay the same.

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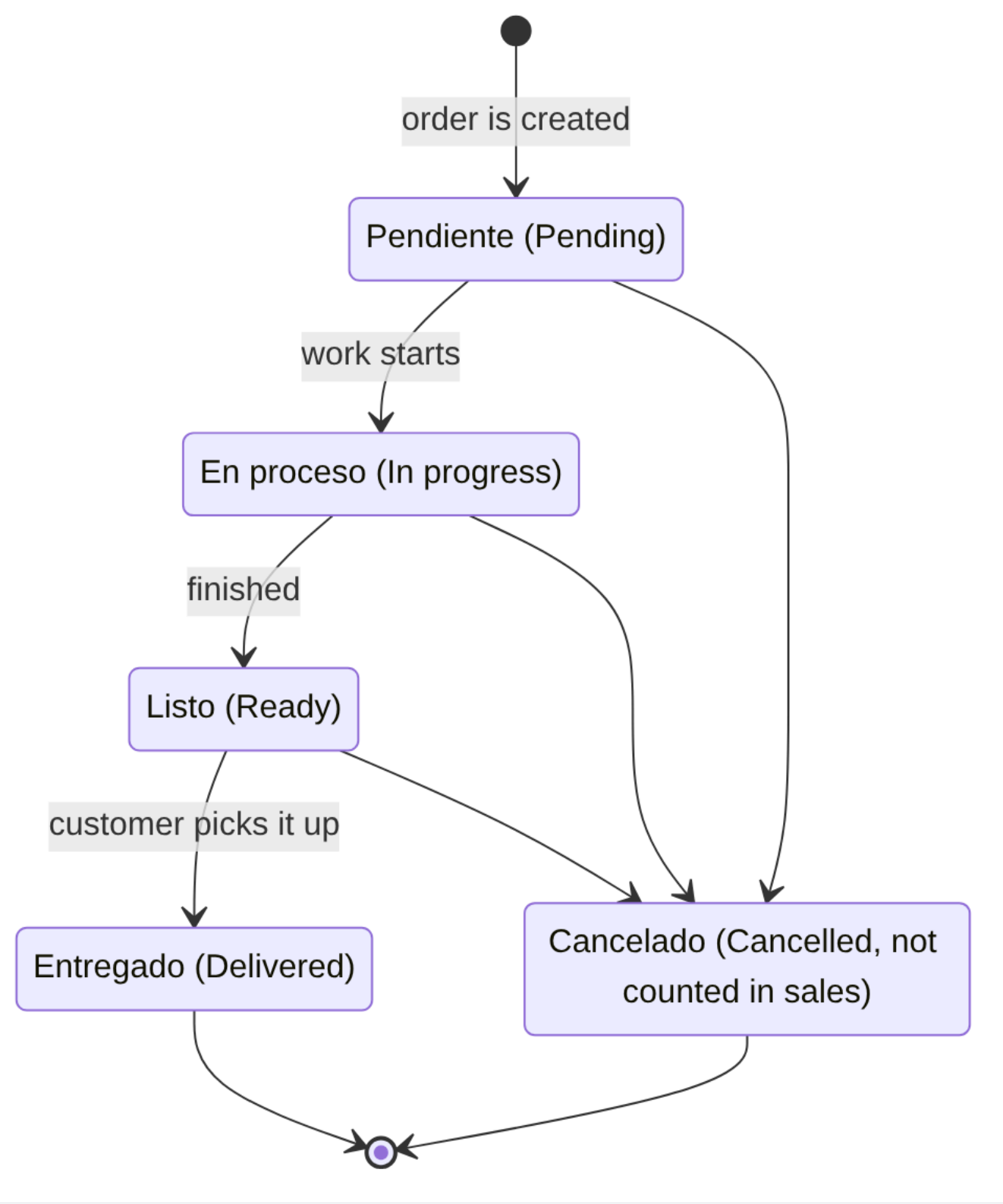
6. Business rules

1. Every order needs a **customer** and **at least one line** with a quantity greater than 0.
2. **Prices are read from the database** on the server. Prices sent by the form are ignored, so nobody can change them from the browser.
3. The order and its lines are saved in a **transaction** (`beginTransaction` / `commit` / `rollBack`): either everything is saved or nothing is.
4. **Cancelled orders are not counted** on the dashboard or in reports.
5. A catalog item **with sales cannot be deleted**. It is set to `activo = 0` instead.
6. Only items with `activo = 1` appear in the new order form.
7. Orders **cannot be edited** after saving. Cancel the order and create a new one.

Order flow


Start
Logged in?
No
Yes
Log in
username and password
New order
Enter customer, phone,
type and notes
Enter quantities of
printing and/or food
System shows
the live total
Save order
Valid data?
No
Yes
Show error
Save order and details
in the database
View order / print ticket
Change status until
Delivered
End

Statuses


order is created
Pendiente (Pending)
work starts
En proceso (In progress)
finished
Listo (Ready)
customer picks it up
Entregado (Delivered)
Cancelado (Cancelled, not counted in sales)

7. Security

Risk	Measure
SQL injection	All queries with user input use **PDO prepared statements**
XSS	All output goes through `e()` (`htmlspecialchars`)
Password exposure	Passwords are stored with `password_hash()` (bcrypt) and checked with `password_verify()`
CSRF	Every POST form carries a token checked with `hash_equals()`
Session fixation	`session_regenerate_id(true)` on login
Unauthorized access	`requiere_login()` on every internal page
SQL script download	`.htaccess` blocks `database/`, `docs/`, `.sql` and `.md` files
Directory listing	`Options -Indexes` in `.htaccess`

Production recommendations:

- Change the `admin` password after installing.
- Set a password for MySQL `root`, or better, create a dedicated user:

```
        CREATE USER 'impresiones'@'localhost' IDENTIFIED BY 'AStrongPassword123';
        GRANT SELECT, INSERT, UPDATE, DELETE ON impresiones_comedor.* TO
'impresiones'@'localhost';
```

  Then update `DB_USER` and `DB_PASS` in `config.php`.
- Run `sudo /opt/lampp/lampp security` to secure XAMPP.

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8. Configuration (`config.php`)

Constant	Default	Description
`NEGOCIO`	Impresiones y Comedor	Business name on screen and ticket
`LUGAR`	Apopa, San Salvador	Address on the ticket
`DB_HOST`	localhost	MySQL server
`DB_NAME`	impresiones_comedor	Database
`DB_USER`	root	MySQL user
`DB_PASS`	*(empty)*	MySQL password
`AREAS`	Impresiones, Comida	Areas (must match the table ENUM)
`CATEGORIAS`	…	Category suggestions in the form
`TIPOS`	Para llevar, Comer aquí, Recoger después	Order types
`ESTADOS`	Pendiente … Cancelado	Valid statuses

The time zone is `America/El_Salvador`. It is also applied to MySQL on connect, so "today" means the same date in PHP and in the database.

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9. Maintenance

9.1 Database backup

```
/opt/lampp/bin/mysqldump -u root impresiones_comedor > backup_$(date +%F).sql
```

Automatic daily backup at 9 p.m. (`crontab -e`):

```
0 21 * * * /opt/lampp/bin/mysqldump -u root impresiones_comedor >
$HOME/backups/impresiones_$(date +\%F).sql
```

9.2 Restore a backup

```
/opt/lampp/bin/mysql -u root impresiones_comedor < backup_2026-10-19.sql
```

9.3 Create a new user

1. Generate the password hash:

```
/opt/lampp/bin/php -r 'echo password_hash("NewPassword123", PASSWORD_DEFAULT), "\n";'
```

2. Insert the user (replace the hash with the output above):

```
INSERT INTO usuarios (usuario, nombre, clave_hash)
VALUES ('cashier1', 'Cashier One', '$2y$10$...output...');
```

9.4 Reset a forgotten password

Generate a hash as in 9.3 and run:

```
UPDATE usuarios SET clave_hash = '$2y$10$...output...' WHERE usuario = 'admin';
```

9.5 Update from GitHub

```
cd /opt/lampp/htdocs/impresiones
git pull
```

(Back up `config.php` first if you changed it.)

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10. Extending the system

I want to…	Change
Add a new status	Add it to `ESTADOS` in `config.php` and an `.est-…` class in `style.css`
Add an order type	Add it to `TIPOS` in `config.php`
Add a third area (e.g. *Stationery*)	Change the `area` ENUM in `catalogo` and add it to `AREAS` and `CATEGORIAS`
Change colors	`:root` variables at the top of `assets/style.css`

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11. Tests performed

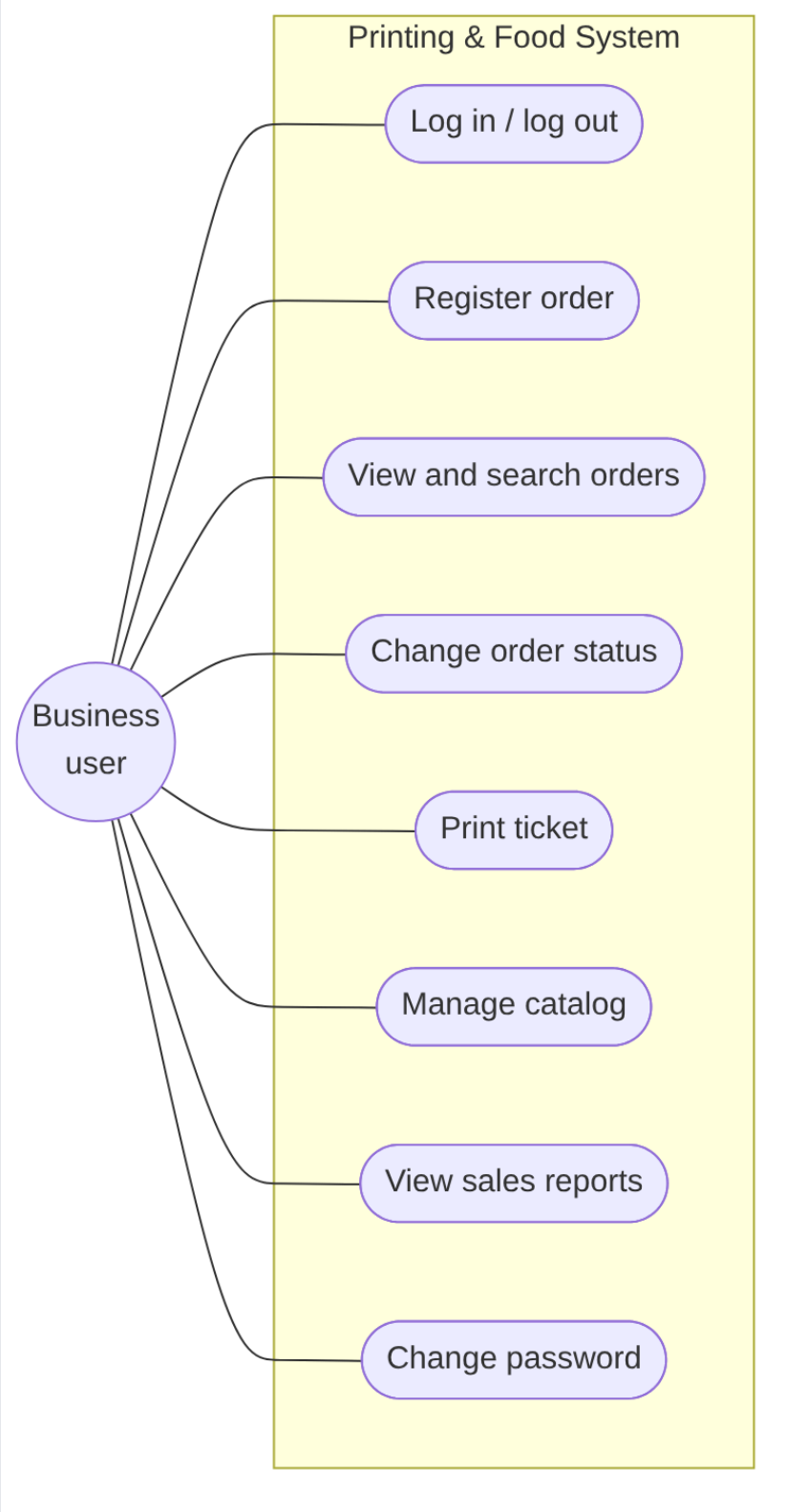
A full automated browser walkthrough checked the following:

Test	Result
Login with wrong credentials shows an error	✅
Correct login opens the dashboard	✅
An internal page without a session redirects to login	✅
Mixed order (printing + food): live total and saved total = $6.50	✅
An order with no items is rejected	✅
Status changes Pending → In progress → Ready	✅
A cancelled order is not counted on the dashboard or in reports	✅
Dashboard totals per area (Printing $3.50, Food $9.00, total $12.50)	✅
Order filters by status and date	✅
Adding and deleting a catalog item	✅
Deleting an item that was already sold deactivates it instead	✅
Report for a date range	✅
Password change and new login	✅
POST request without a CSRF token is rejected (HTTP 400)	✅
No PHP or JavaScript errors or warnings	✅

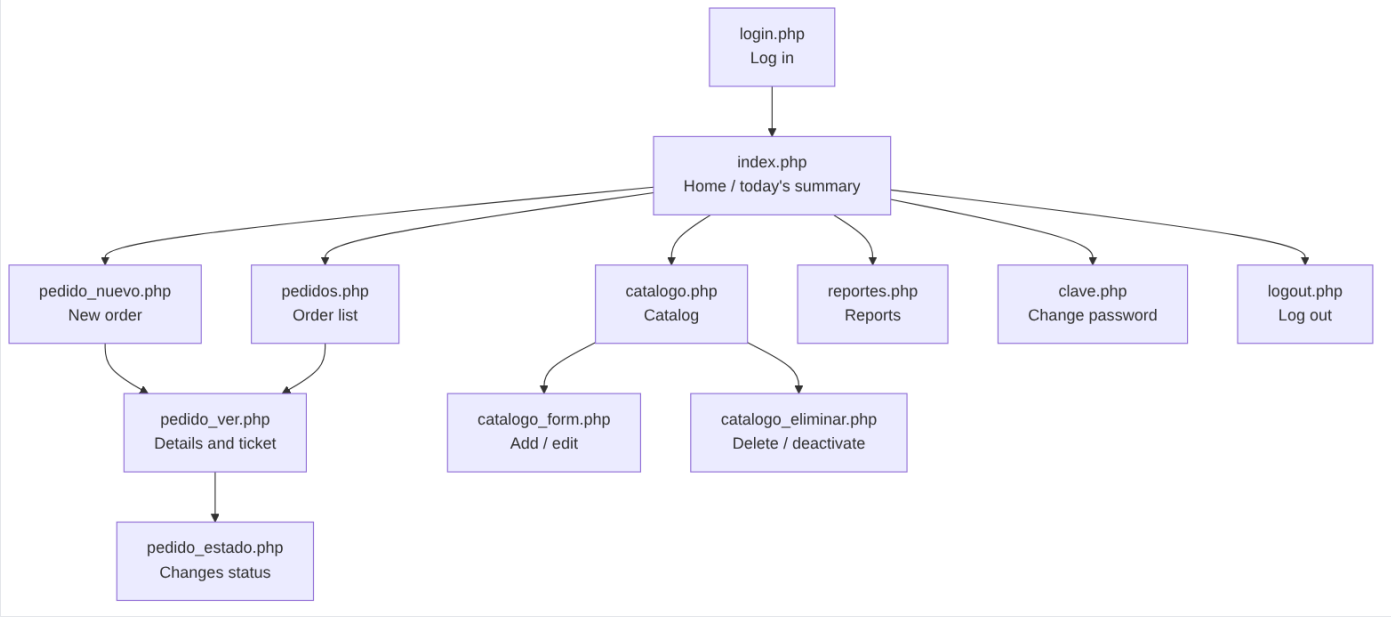
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12. More diagrams

- Use cases:


Printing & Food System
Log in / log out
Register order
View and search orders
Change order status
Business user
Print ticket
Manage catalog
View sales reports
Change password

- Site map:



- All diagrams with source code: see diagrams.md